ch20 Electrostatics

Items:

Q11
Q8
Q9
Q7

Q11

11. Figure 11.1 shows two identical small metal spheres X and Y suspended by insulating threads of the sa length. Each sphere has a mass of 1.0×10^{-5} kg and each carries a positive charge of 3.1 nC (1 nC = 10^{-9} The separation d of the spheres is 10 cm. The size of the spheres is negligible compared with t separation, therefore they can be treated as point charges.

Take $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ N m^2 C^{-2}. ($g = 9.81$ m s^{-2})

Figure 11.1

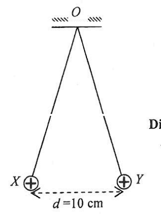

Diagram NOT drawn to scale

(a) Find the angle between the threads. (3 mar

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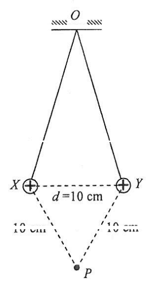
(b) Point P is vertically below the fixed point O and it is 10 cm from each sphere.

(i) Indicate the direction of the resultant electric field at P due to these two charged spheres.(1 ma

Figure 11.2



*(ii) Calculate the electric potential at point P. The electric potential at infinity is taken to be zero. (2 mark

(iii) A neutral metal sphere of finite size is now placed at P. State whether the separation d wou increase, decrease or remain unchanged due to the presence of this sphere. (1 marl

Separation d ....

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Pape* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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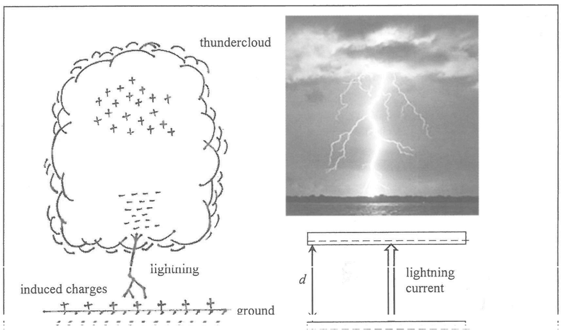
Q8

8. Read the following passage about **lightning** and answer the questions that follow.



Lightning occurs when charges accumulate in the clouds to such an extent that the electric field in the atmosphere is strong enough to cause the air to lose its insulating properties. The threshold electric field for 'breakdown' to occur is about 3×10^5 V m^{-1} above which electrons or ions in the atmosphere can pass through the air between clouds and the ground or between clouds and clouds. The peak current of a typical lightning bolt can reach about 30000 A. How the charges are separated and accumulated in the clouds is not fully understood yet. In most cases, negative charges are at the base of the cloud and positive charges are induced on the ground.

(a) (i) What is the meaning of 'breakdown' in the passage ? (1 mark)

*(ii) The thundercloud's base and the ground can be modeled as two parallel plates with opposite charges. If the negative charges distributed at the cloud's base are about d = 2 km from the ground, find the potential difference between the cloud and the ground when the electric field in the atmosphere just reaches the threshold of 'breakdown'. (2 marks)

A lightning detector having a small square coil inside is placed at point O which is 1.5 km from the lightning bolt. The coil and the lightning's direction are in the same vertical plane as shown. Assume that the lightning current flows vertically upwards to the thundercloud from the ground.

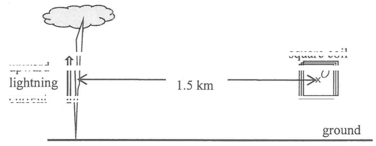


Diagram **NOT** drawn to scale

(b) (i) State the direction of the magnetic field (to the left / to the right / into paper / out of paper) produced at point O by the lightning current. Estimate the magnetic field strength's peak value at O. (3 marks)

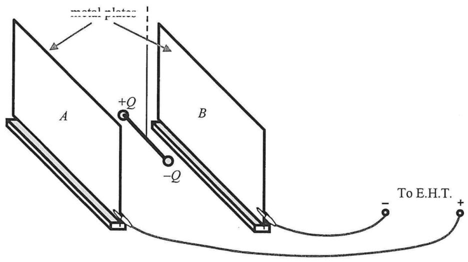
(ii) Explain why within the very short duration of lightning an induced current first flows in the coil in a certain direction and then reverses. Your answer should include the directions of the induced current in the coil. (2 marks)

(iii) Among the physical quantities related to lightning, **electric field in the atmosphere**, **lightning current** and **magnetic field due to lightning**, suggest which one can be monitored so as to give fore-warning of lightning. Explain your choice. (2 marks)

Q9

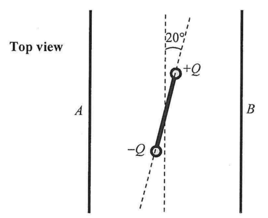
9. Two small metal spheres are attached to the ends of an insulating rod of length 5.0 cm. They carry charges $+Q$ and $-Q$ respectively of equal magnitude as shown in Figure 9.1. The insulating rod is suspended horizontally between two parallel metal plates, A and B, which are connected to an E.H.T. (extra high tension) supply.

Figure 9.1



The rod is parallel to the metal plates when the E.H.T. is off. After the E.H.T. is switched on, an electric field is set up between the plates and the rod is twisted by an angle of 20° as shown in Figure 9.2.

Figure 9.2



(a) On Figure 9.2, sketch the electric field lines due to the potential difference across the plates. (2 marks)

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(b) The potential difference across A and B is 5.0 kV and the separation between the metal plates is 10 cm. The force due to the electric field acting on each sphere is 2.0×10^{-5} N, find

(i) the moment acting on the rod as shown in Figure 9.2 due to the electric forces on the charged spheres. (2 marks)

*(ii) the strength of the electric field E due to the potential difference across the metal plates. (2 marks)

(iii) the magnitude of the charge Q on the spheres. (2 marks)

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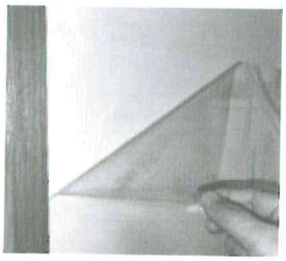
Q7

7. Read the following passage about **electrostatic adhesive wallpaper** and answer the questions that follow.

Electrostatic adhesive wallpaper is a wall covering that uses **static electricity** for adhesion, without the need for traditional glues or pastes. The wallpaper is typically made from a material that can retain electric charges on its surface and is charged through friction during production, allowing it to adhere directly to smooth, clean surfaces like painted walls or glass.



The wallpaper can be easily applied or removed. After removal, the wallpaper does not leave residue or damage the underlying surface. Therefore, the wallpaper is especially suitable for quick updates and temporary setups. Another advantage of the wallpaper is its reusability. It preserves its adhesive properties after being removed and can be reapplied.

(a) Describe how the electrostatic adhesive wallpaper can adhere to a painted wall that is initially neutral. (2 marks)

(b) Explain briefly why the wallpaper preserves its adhesive properties after being removed. (2 marks)

(c) The wallpaper **does not** adhere well to the wall in a humid environment. Explain briefly. (1 mark)

Answers written in the margins will not be marked.